

# Optimal Block Time for AMM Liquidity Providers under Jump-Diffusion Prices

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# The case for faster blocks

# Two live conversations about how fast a chain should be

## Arbitrum Research — “The power of faster blocks” (Felten, May 2024)

“Arbitrum’s 250 millisecond block time leads to **65% lower arbitrage loss** compared to a 2 second block time.”

[research.arbitrum.io/t/the-power-of-faster-blocks/9609](https://research.arbitrum.io/t/the-power-of-faster-blocks/9609) ↗

## EIP-7782 “Reduce Block Latency” — 12 s → 6 s, drafted for Glamsterdam

“More frequent blocks **decrease LVR** (Loss Versus Rebalancing), which improves the economics for liquidity providers. More liquidity means lower spreads.”

[eips.ethereum.org/EIPS/eip-7782](https://eips.ethereum.org/EIPS/eip-7782) ↗

Both rest on the same result. Both, on the numbers below, overstate the gain — by roughly a factor of two.

# Where that argument comes from

Passive liquidity providers on an AMM lose to arbitrageurs whenever the external price moves — **loss-versus-rebalancing** (LVR), Milionis et al. (2022).

- Under geometric Brownian motion, a constant-product pool loses at the rate  $\sigma^2 V/8$ , independent of the block time.
- Add a swap fee  $\gamma$  and Poisson blocks of mean  $\Delta t$  (Milionis et al., 2023): the rate is multiplied by  $F(\kappa)$ ,  $\kappa = \gamma/(\sigma\sqrt{\Delta t})$  — the stationary probability that an arriving block offers a *profitable* trade.
- $F(\kappa) \sim \sigma\sqrt{\Delta t}/(\sqrt{2}\gamma) \rightarrow 0$  as  $\Delta t \rightarrow 0$ . That  $\sqrt{\Delta t}$  is where Arbitrum's 65% comes from, and the  $\sim 29\%$  implied by halving Ethereum's slot; and it says shorter blocks mean less LVR, all the way down.

## The deployed prescription

“Shorten blocks.” Solana at 400 ms, Ethereum L2s at 2 s, app-chains chasing 50 ms — with LP welfare among the stated reasons.

# The primitive that prescription assumes

The whole argument rests on the reference price being **continuous**. A diffusive gap has size  $O(\sigma\sqrt{\Delta t})$ , so halving the block halves what the arbitrageur finds.

Crypto prices are not continuous:

- Scaillet et al. (2020): frequent, clustered jumps in BTC at high frequency.
- Saef et al. (2024): BTC jumps on  $\sim 58\%$ , ETH on  $\sim 32\%$  of trading days.
- My own ETH/USDT calibration:  $\lambda = 283$  jumps/yr.

**A jump does not care about  $\Delta t$**

A jump of size  $J$  opens a price gap of size  $J$  whatever the block schedule. Faster blocks clear each gap *sooner* — they do not make it *smaller*.

# This talk in one line

## Result

Under jump-diffusion the LVR rate splits into a diffusion channel the block schedule governs and a jump channel it cannot touch. For symmetric jump laws the second is an **exact lower bound** — a floor no block speed breaches.

What follows:

1. A zero-fee decomposition (Theorem 1).
2. With fees and blocks: a separated rate plus a two-sided bounded remainder (Theorem 2, Proposition 1).
3. The floor, exactly, and the  $\sqrt{\Delta t}$  approach to it (Theorem 3, Corollary 1).
4. The LP-side planner's optimum: a cubic, invariant in pool size and in every jump parameter (Theorem 4).
5. ETH/USDT calibration: 471 bp/yr at 12 s against a floor of 125, optimum near 8 s — and the two opening claims become **35%** and **14%**.

Two channels, and a floor

# Setup

Milionis et al. with **one primitive changed**: the reference price jumps.

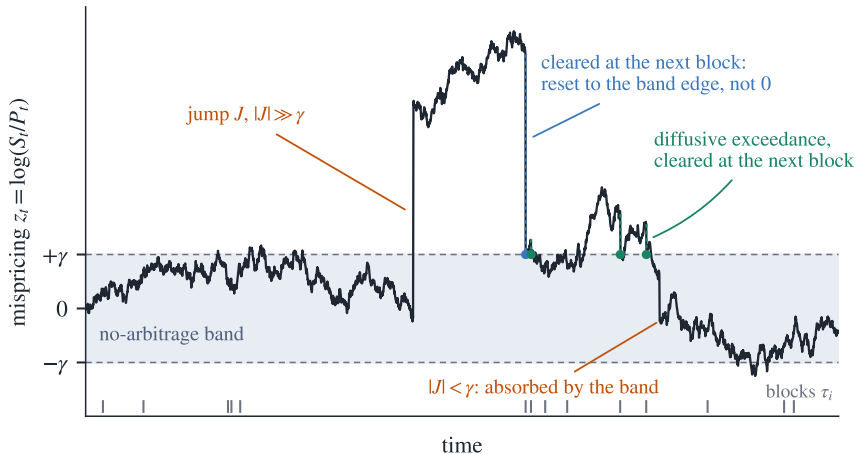
$$\frac{dS_t}{S_{t-}} = \mu dt + \sigma dW_t + (e^{J_t} - 1) dN_t, \quad N \text{ Poisson}(\lambda), \quad J \sim \nu.$$

- $\nu$  finite activity with  $\int (e^{j/2} - 1)^2 \nu(dj) < \infty$ ; Merton  $J \sim \mathcal{N}(m, \delta^2)$  the running instance.
- CPMM,  $x_t y_t = L$ , pool value  $V(P) = 2\sqrt{LP}$  — concave, and that concavity *is* the loss.
- Fee  $\gamma$  in log-price units  $\Rightarrow$  no-arbitrage band  $\{|\log S - \log P| \leq \gamma\}$ ; blocks Poisson of rate  $1/\Delta t$ .
- $\text{LVR}(T) = \int_0^T V'(P_{s-}) dP_s - [V(P_T) - V(P_0)]$ ; rate  $\ell = \mathbb{E}_t[d\text{LVR}_t]/dt$ .

## What is general and what is not

Results need **symmetry** of  $\nu$ , not Gaussianity. Theorems 1, 2, 4 hold for arbitrary  $\nu$ ; Proposition 1 and Theorem 3 need symmetry.

# What the model looks like



The pool-reference log gap  $z_t = \log(S_t/P_t)$ . Diffusive exceedances (teal) scale with  $\sigma\sqrt{\Delta t}$ ; a jump (orange) opens a  $\Delta t$ -independent gap.

# Theorem 1 — the zero-free decomposition

In the continuous-arbitrage limit ( $\gamma = 0$ ,  $\Delta t \rightarrow 0$ ),  $P_t \equiv S_t$  and LVR is the Itô–Lévy remainder of  $V(S_t)$ :

$$\ell = \underbrace{\frac{\sigma^2}{8} V(S_t)}_{\text{diffusion-LVR}} + \underbrace{\frac{\lambda}{2} V(S_t) \mathbb{E}[(e^{J/2} - 1)^2]}_{\text{jump-LVR}},$$

and for Merton  $\mathbb{E}[(e^{J/2} - 1)^2] = e^{m+\delta^2/2} - 2e^{m/2+\delta^2/8} + 1 \geq 0$ , zero iff  $J \equiv 0$ .

## Two observations

The jump term carries **no**  $\Delta t$ . At  $\gamma = 0$  the arbitrageur clears any gap in the next block, so refining the schedule only redistributes *when* the loss lands.

## Measure

The drift cancels between the two terms, so no normalisation of  $\mu$  is needed — but with jumps the rate is *not* measure-independent, so this is a physical-measure statement and calibration is against realised prices.

# The mispricing is a persistent state

Write  $z_t := \log(S_t/P_t)$ . At a block the arbitrageur trades iff  $|z| > \gamma$  and moves the pool to the *near band edge*:  $z \mapsto \text{bound}(z, -\gamma, \gamma)$  — not to 0.

$$z_t = \left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma W_t + \underbrace{\sum_{k: T_k \leq t} J_k}_{\text{exogenous, rate } \lambda} + \underbrace{\sum_{i: \tau_i \leq t} \mathcal{J}_i}_{\text{endogenous, rate } 1/\Delta t}$$

## The asymmetry every result comes from

Two point processes drive  $z$ : price jumps push it **away** from the band at a rate the protocol does not choose; block-triggered resets push it **back** at a rate the protocol does choose.

Per block, loss  $h(\xi) = \frac{\gamma}{2}(e^{\xi/2} - 1)^2$  with  $\xi = \text{sign}(z)(|z| - \gamma)_+$ , and  $\ell(\Delta t) = \frac{1}{\Delta t} \mathbb{E}_\pi[h(\xi)]$  against the stationary law  $\pi$ .

## The two channels, separately

**Diffusion (Lemma 1).** At  $\lambda = 0$  the stationary law is uniform on the band with exponential tails, so

$$\ell_{\text{diff}} = \frac{\sigma^2}{8} V \cdot F(\kappa), \quad F(\kappa) = \frac{1}{1 + \sqrt{2} \kappa}$$

with  $\kappa = \gamma/(\sigma\sqrt{\Delta t})$ ;  $F(\kappa) = \mathbb{P}_\pi(|z| > \gamma)$ , strictly decreasing,  $F(\kappa) \rightarrow 0$ .

**Jumps (Lemma 2).** The jump contribution is  $\lambda V G_\nu(\gamma)$  with

$$G_\nu(\gamma) = \frac{1}{8} \mathbb{E}_\nu[(|J| - \gamma)^2 \mathbf{1}\{|J| > \gamma\}],$$

a functional of  $\nu$  alone. For symmetric Merton,  $G = \frac{\delta^2}{8} \Psi(\gamma/\delta)$ .

### The fee discount is different on each channel

Both channels are trimmed by the fee — but by  $F(\gamma/\sigma\sqrt{\Delta t})$  on one and by  $\Psi(\gamma/\delta)$  on the other.

**Only the first contains  $\Delta t$ ,** because the size of a jump is set by  $\delta$ , not by how long the block lasted.

## Theorem 2 — the separated rate

$$\ell(\Delta t) = \ell_0(\Delta t) + \mathcal{E}(\Delta t), \quad \ell_0(\Delta t) := \frac{\sigma^2}{8} V \cdot F\left(\frac{\gamma}{\sigma\sqrt{\Delta t}}\right) + \lambda V \cdot G_\nu(\gamma).$$

This is a *definition* of  $\mathcal{E}$  and carries no content until  $\mathcal{E}$  is controlled.

**Proposition 1** bounds it on both sides — six signed terms, no order symbols, under a stated mixing condition  $n_* \leq 2(1 + \eta^2)$  checked numerically over every block time, volatility and fee tier considered:

- interaction ( $|M| \leq \gamma$  pathwise), block clock, single-jump thinning, multi-jump, curvature residual (exact, closed form), stationary law.
- At the calibration:  $\mathcal{E}(\Delta t) \in [-0.653, +0.849]$  bp/yr at 12 s, never above 0.29% of the rate over 50 ms–12 s. Quadrature puts the realised remainder at +0.165 bp/yr, inside the bound and of the predicted sign.

## Theorem 3 — an exact floor

### No approximation, no remainder

Let  $\nu$  be symmetric and satisfy  $\mathbb{E}[g(\sum_1^k J)] \geq k \mathbb{E}[g(J)]$  for every  $k$  (Merton with  $m = 0$  does).  
Then for every  $\Delta t > 0$

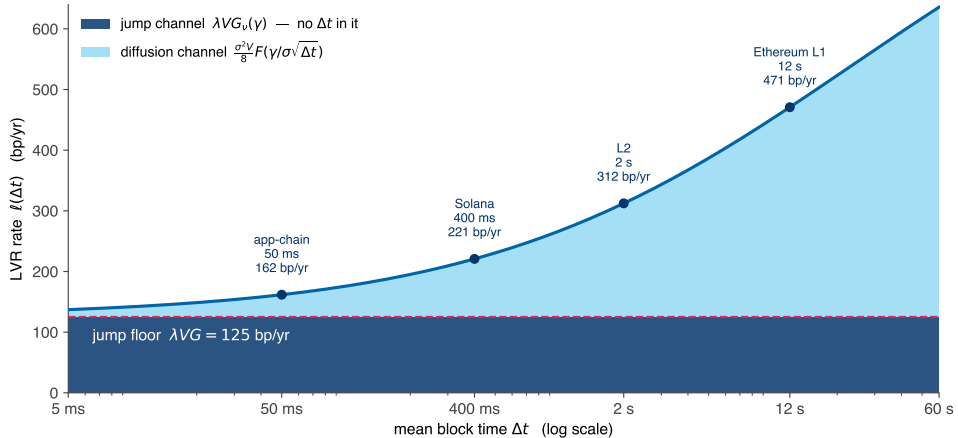
$$\ell(\Delta t) \geq \lambda V G_\nu(\gamma) = \inf_{\Delta t' > 0} \ell_0(\Delta t') > 0.$$

- Proved from convexity of the quadratic surrogate — it never touches the stationary law, which for  $\lambda > 0$  has no closed form. That is what makes it available, and why it does *not* inherit Proposition 1's mixing condition.
- **Corollary 1.**  $\ell_0$  is strictly increasing and the floor is approached only as  $\sqrt{\Delta t}$ : halving the block time removes at most 29% of the diffusion term.

### Keep two limits apart

$\Delta t \rightarrow 0$  shortens the interval between blocks. The jump law is exogenous to the schedule, so it does *not* shrink  $J$ .

# The two channels, drawn



$\ell_0(\Delta t)$  at the ETH/USDT calibration. The dark band is constant in  $\Delta t$ ; only the pale band responds to the block schedule.

Numbers, and the planner's optimum

# Calibration, and the rate across deployed block times

Binance ETH/USDT 5-minute closes, Jan 2020 – Jun 2026 (2,373 days, 682,946 bars):  $\sigma = 0.8156$  from bipower variation;  $\lambda = 283.3/\text{yr}$  and  $\delta = 0.0192$  from Lee–Mykland at  $\alpha = 1\%$ ;  $m = 0$  ( $\hat{m} = -0.0012$ , 99% CI contains zero). Modelling choices:  $\gamma = 5$  bp (Uniswap 0.05%),  $V = \$1\text{M}$ .

| Chain        | $\Delta t$ | $\ell$ (bp/yr) | diff. share |
|--------------|------------|----------------|-------------|
| Ethereum L1  | 12 s       | 471            | 73%         |
| Base / OP L2 | 2 s        | 312            | 60%         |
| Solana       | 400 ms     | 221            | 43%         |
| Arbitrum     | 250 ms     | 203            | 38%         |
| App-chain    | 50 ms      | 162            | 23%         |
| Jump floor   | —          | 125            | 0%          |

- Ceiling  $\sigma^2 V/8 = 832$  bp/yr; floor  $\lambda VG = 125$ .
- A  $240\times$  cut,  $12\text{ s} \rightarrow 50\text{ ms}$ , removes only 89% of the diffusion term and leaves the rate 29% above the floor.
- The channels are equal at  $\Delta t_{\times} \approx 0.75\text{ s}$ ; the diffusion term reaches a tenth of the floor only at  $\Delta t \approx 5.5\text{ ms}$ .

## Theorem 4 — netting LVR against what a block costs

Fixed per-block work gives a cost rate  $R(\Delta t) = r/\Delta t$ ; a pool inherits  $c/\Delta t$  with  $c = r V/V_{\text{chain}}$ . The LP-side planner minimises  $W(\Delta t) = \ell_0(\Delta t) + c/\Delta t$ :

$$\eta^{\text{opt}}(1 + \eta^{\text{opt}})^2 = \frac{V\gamma^2}{8c}, \quad \Delta t^{\text{opt}} = \frac{2\gamma^2}{\sigma^2(\eta^{\text{opt}})^2},$$

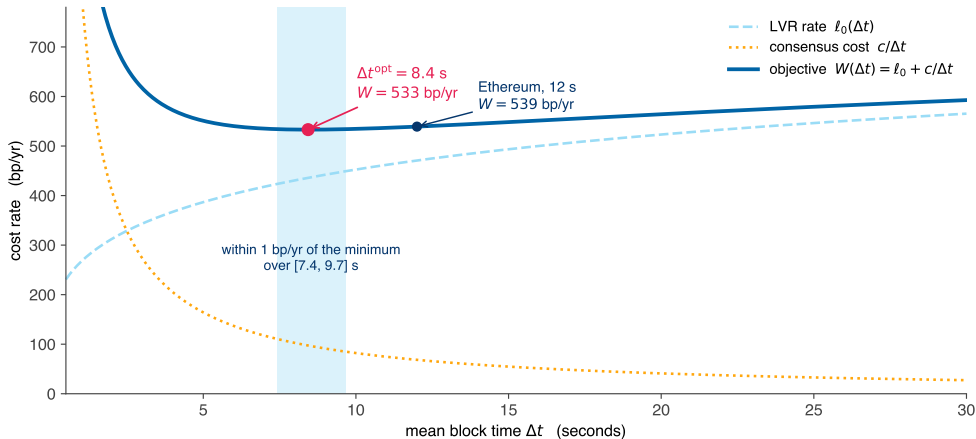
a cubic with a unique positive root, closed form by Cardano.

- $\partial\Delta t^{\text{opt}}/\partial\lambda = \partial\Delta t^{\text{opt}}/\partial m = \partial\Delta t^{\text{opt}}/\partial\delta = 0$ : **the jump parameters do not enter the first-order condition.**
- $\partial\Delta t^{\text{opt}}/\partial V = 0$ , exactly —  $V$  cancels, since  $c \propto V$ . The optimum is a property of the chain and the fee tier, not of the pool.

### What the remainder does to it

That invariance is a property of  $\ell_0$ ;  $\mathcal{E}$  carries both  $\lambda$  and  $\Delta t$ . Bounding it directly puts any exact minimiser in  $[7.3, 9.9]$  s — the range over which  $W$  is within 1 bp/yr of its minimum anyway.

# The objective, and its shallow optimum



The optimum is shallow: 539  $\rightarrow$  533 bp/yr. Across plausible  $c$  it runs 0.2–15 s; the jump parameters move it not at all.

What this means, and what it does not

# Policy

## Back to the two proposals

Arbitrum, 2 s  $\rightarrow$  250 ms:  $\sqrt{\Delta t}$  gives 65%, the model gives 312  $\rightarrow$  203 bp/yr, a 35% cut.

EIP-7782, 12 s  $\rightarrow$  6 s:  $\sqrt{\Delta t}$  gives 29%, the model gives 471  $\rightarrow$  403 bp/yr, a 14% cut. Both price the diffusion channel as the whole loss.

- Shortening blocks can at most close  $\ell(\Delta t) - \lambda VG$  — and only as  $\sqrt{\Delta t}$ .
- The diffusion channel stays material into the low-millisecond range. **So at every deployed block time, what limits block-time policy is not the floor — it is the cost of consensus.**
- Once the floor does bind, the remaining LP-side levers are not the schedule:
  - ▶ fee-tier design, since the binding ratio is  $\gamma/\delta$ ;
  - ▶ batch auctions, which remove the per-block arbitrage window at source;
  - ▶ LVR-rebate designs, e.g. auction-managed AMMs routing pool-management revenue back to LPs.

## Key takeaway

Jumps change what block time can *achieve*, not what it should *be*.

# Conclusion

- **125 bp/yr** — the jump floor. Exact for symmetric  $\nu$ , at every block time, with no remainder.
- **471 bp/yr** at Ethereum's 12 s, 73% of it in the channel the schedule governs — and that channel yields only as  $\sqrt{\Delta t}$ .
- **8.4 s** — the LP-side optimum. A cubic; invariant in pool size exactly, and in  $(\lambda, m, \delta)$  below the objective's flatness.
- So both proposals I opened with are right in direction, and about twice too optimistic.

## Key takeaway

The rate splits into a channel the block schedule governs and one it cannot touch. The floor is genuine but distant: what binds at deployed block times is the cost of a block, and the lever against the floor is  $\gamma/\delta$ .

This is an LP-side input to a larger welfare problem, not a prescription — limitations on the backup slides. Extended version, with all proofs: [arXiv:2608.30321](https://arxiv.org/abs/2608.30321) ↗.

# Questions and discussion.

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Backup

# Limitations

- **Objective is LP-LVR-only.** A full welfare treatment must also weigh MEV redistribution, finality, censorship resistance and oracle freshness — all of which could push  $\Delta t^{\text{opt}}$  back up.
- **Consensus cost.** Issuance is a chosen compensation policy, not a measured resource cost; it overstates where it subsidises growth and understates where MEV and fees fund validators.
- **Full-range pool.** Most LVR is paid on concentrated liquidity, where the two effects on  $G$  run in opposite directions — so the full-range floor is not simply a lower bound there.
- **Idealised arbitrageur:** unlimited capital, zero latency, continuous sight of  $S_t$  — weakest exactly in the sub-second regime the floor argument concerns.
- **Poisson blocks**, inherited for tractability. Deterministic slots give the same shape with  $C = -\zeta(1/2)/\sqrt{2\pi} \approx 0.583$  in place of  $1/\sqrt{2}$ , and move  $\Delta t^{\text{opt}}$  by 5%.

## Fees split LVR — they do not reduce it

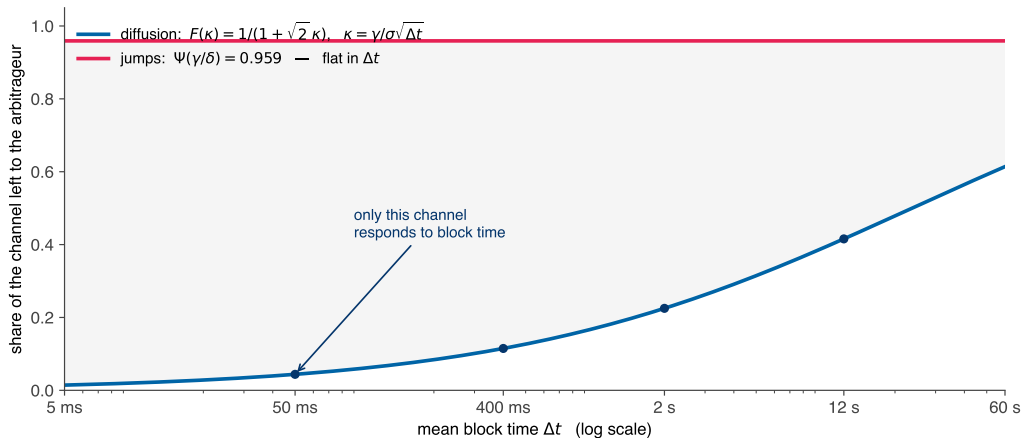
In the accounting of Milionis et al., arbitrage profit and fee income sum to the frictionless LVR of Theorem 1. So  $\ell(\Delta t)$  is the arbitrage half: LP loss *net* of the fees LPs get back.

- Diffusion channel: the split is  $F(\kappa) : 1 - F(\kappa)$ , and  $F(\kappa) \rightarrow 0$ . Shorter blocks *do* return diffusion-LVR to LPs as fee income.
- Jump channel: the split is  $\Psi(\gamma/\delta) : 1 - \Psi(\gamma/\delta)$  — a function of  $\gamma/\delta$  alone, with no  $\Delta t$  in it.
- At the calibration  $\gamma/\delta = 0.026$ , so fees recover 4.1% of jump-LVR. The rest is the floor.

### Key takeaway

The lever against the floor is the ratio of fee to jump size — fee-tier design — not the block schedule.

# Two fee discounts, one lever



The share of each channel left to the arbitrageur.  $F$  collapses as blocks shorten;  $\Psi(\gamma/\delta)$  does not move.

# Which regime a chain is in matters more than its slot

$\ell$  in bp/yr at  $\Delta t = 12$  s, diffusion share in brackets:

| $\sigma \setminus \lambda$ | 71/yr       | 283/yr      | 1,133/yr    | 4,533/yr    |
|----------------------------|-------------|-------------|-------------|-------------|
| 0.30 (calm)                | 55 (43%)    | 148 (16%)   | 524 (4%)    | 2,027 (1%)  |
| 0.816 (baseline)           | 377 (92%)   | 471 (73%)   | 847 (41%)   | 2,349 (15%) |
| 1.50 (stress)              | 1,625 (98%) | 1,719 (93%) | 2,095 (76%) | 3,597 (44%) |

- The *level* spans a factor of 65; the *share* runs from 1% to 98% and moves the *other* way in  $\lambda$ .
- The share is what decides whether shortening blocks helps at all. At  $\lambda = 283/\text{yr}$  it crosses 50% at  $\sigma = 0.55$ .

## Key takeaway

Any policy claim about 12-second blocks is conditional on the volatility-and-jump-intensity regime, not on a single point estimate.

# What actually moves the optimum

| $c (\times 10^{-5} \text{ USD/block})$ | $\Delta t^{\text{opt}}$ |
|----------------------------------------|-------------------------|
| 500                                    | 15.4 s                  |
| 260 (baseline)                         | 8.4 s                   |
| 90                                     | 3.4 s                   |
| 30                                     | 1.4 s                   |
| 2                                      | 0.20 s                  |

| $\sigma$      | $\Delta t^{\text{opt}}$ |
|---------------|-------------------------|
| 0.30 (calm)   | 62.3 s                  |
| 0.816         | 8.4 s                   |
| 1.50 (stress) | 2.5 s                   |

- Nearly two orders of magnitude across the plausible range of the cost calibration; volatility moves it comparably, since  $\Delta t^{\text{opt}} \propto \gamma^2 / \sigma^2$  at fixed  $\eta$ .
- The jump parameters do not move it at all.

## Key takeaway

Under an LP-LVR-only objective, Ethereum's 12 s is somewhat too slow and sub-second targets overshoot — but the gain over the status quo is under 2%, and the answer is driven by  $c$ ,  $\sigma$ ,  $\gamma$ .

# The reset-to-zero shortcut

A common simplification is to assume a trade resets the mispricing  $z$  to 0 rather than to the band edge  $\pm\gamma$  — i.e. to use  $\Psi(\kappa)$  in place of  $P_{\text{trade}} = F(\kappa)$ .

## It is not a small error

It understates the diffusion term by  $2.7\times$  at 12 s,  $76\times$  at 2 s, and  $4\times 10^7$  at 400 ms — because  $\Psi$  has a Gaussian tail where  $F$  decays only algebraically.

The persistence of the mispricing (Lemma 1) is what makes the diffusion channel decay as  $\sqrt{\Delta t}$  rather than vanish, and so it is also what keeps that channel material down into the millisecond range.